ModelIt!

Teacher's Guide

Can We Engineer the Amazon Back to Life?

Modeling Rainforest Restoration and Ecological Tipping Points

Quick Reference

Grade Level	Grade 10L2
NGSS Standard	HS-LS2-7, HS-ESS2-7
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-LS2-7, HS-ESS2-7

Design, evaluate, and refine a solution for reducing the impacts of human activities on the environment and biodiversity. Construct an argument based on evidence about the simultaneous coevolution of Earth's systems and life on Earth.

Three-Dimensional Learning

Science Practice	**Designing Solutions** — Students design evidence-based restoration plans using computational model data to optimize species selection, spatial arrangement, and timeline milestones.
Disciplinary Core Idea	**LS2.C Ecosystem Dynamics, Functioning, and Resilience / ESS2.A Earth Materials and Systems** — Ecosystems maintain function through interconnected biotic and abiotic processes; disruption of key components can trigger cascading failure and regime shifts.
Crosscutting Concept	**Stability and Change** — Students model how the Amazon ecosystem maintains stability through self-reinforcing feedback loops and identify the threshold conditions where stability is lost.

CAST Testing Alignment

MULTIPLE CHOICE:

The Amazon has lost 17% of its original forest cover. Based on your model, what is the most urgent action to prevent crossing the estimated 20-25% tipping point?

CONSTRUCTED RESPONSE:

Using your model, explain why simply replanting trees in deforested areas is not sufficient to restore the Amazon's ecological function. Reference Soil Nutrient Recovery, Water Cycle Restoration, and Biodiversity Return in your explanation.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don’t need to teach all of this — it’s here so you can answer student questions accurately.

The Amazon rainforest — 5.5 million square kilometers spanning nine countries — is the most biodiverse ecosystem on Earth, home to 10% of all known species, storing 150-200 billion tons of carbon, and generating half of its own rainfall. It is also approaching a catastrophic tipping point that could transform it from the world's greatest carbon sink into a massive carbon source.

The Amazon's Self-Sustaining Water Cycle

Unlike most forests, the Amazon creates its own weather. Trees pump water from the soil through their roots and release it as water vapor through their leaves — a process called evapotranspiration. A single large Amazon tree can release 1,000 liters of water per day. This moisture forms clouds that travel westward, generating rainfall that sustains the forest downwind. Approximately 50% of the Amazon's rainfall is recycled moisture from the forest itself. This means the forest needs itself to survive — remove enough trees and the remaining forest doesn't get enough rain to sustain itself.

The Tipping Point

Scientists estimate the Amazon's tipping point lies at 20-25% total deforestation — the level at which the forest can no longer generate sufficient rainfall to sustain itself. As of 2024, approximately 17% of the original Amazon has been cleared. At current deforestation rates of 10,000 square kilometers per year, the tipping point could be reached within 15-25 years. Once crossed, models predict a cascade: reduced rainfall causes tree death, which reduces evapotranspiration, which further reduces rainfall, converting most of the eastern and southern Amazon to degraded savanna within decades.

The Challenge of Restoration

Replanting trees is necessary but woefully insufficient for restoring a tropical rainforest. The Amazon's biodiversity includes over 80,000 plant species with intricate ecological relationships — mycorrhizal fungal networks that connect tree roots, specific pollinators for each plant species, seed dispersers that distribute species across the landscape. Soil in deforested areas loses its organic layer, fungal networks, and nutrient cycling capacity within years. Rebuilding these foundations takes 15-50 years under optimal conditions. Full canopy forest with specialist interior species requires 100-200+ years.

Success Stories and Lessons

Despite the daunting challenges, some restoration projects show promise. Costa Rica reversed decades of deforestation, going from 21% to 60% forest cover in 30 years through payments for ecosystem services. Atlantic Forest restoration in Brazil has replanted millions of hectares using natural regeneration assisted by seed dispersal. Key lessons: restoration near intact forest succeeds faster (seed sources and wildlife corridors), mixed species plantings outperform monocultures, and involving local and indigenous communities is essential for long-term success.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify seven components of the Amazon restoration system: Deforestation Rate, Reforestation Effort, Soil Nutrient Recovery, Water Cycle Restoration, Biodiversity Return, Carbon Sequestration, and Tipping Point Threshold — with Deforestation Rate and Reforestation Effort as external drivers.

E **Establish Relationships**

Students determine that Deforestation Rate pushes the system toward the tipping point while Reforestation Effort pulls it back. Recovery depends on Soil Nutrient Recovery enabling Water Cycle Restoration, which enables Biodiversity Return, which enables full Carbon Sequestration. The timescales are decades to centuries.

V **Visualize & Model**

Students build a computational model showing the race between deforestation and restoration, with the tipping point as the critical threshold that determines the Amazon's fate.

E **Evaluate Outcomes**

Students run continued deforestation, aggressive restoration, and partial intervention scenarios to determine which strategies can prevent tipping point crossing and what recovery timelines look like.

R **Revise & Extend**

Students add Fire Vulnerability, Indigenous Land Management, or Economic Pressure to explore how additional factors affect restoration feasibility and tipping point dynamics.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"The Amazon generates half of its own rain. It is the only forest on Earth that literally creates the weather it needs to survive. And we are cutting it down."*

Watch For

Show satellite comparison of Amazon deforestation over 20 years. Ask: What happens when a forest that makes its own rain loses too many trees to make rain?

Slide 2: Learning Objectives

3 min

Say: *"Today you're going to determine whether the Amazon can be saved — or whether we're already too late."*

Watch For

Have students read objectives. Pre-teach 'tipping point threshold' and 'ecological succession.' Quick-write: How long do you think it takes to regrow a rainforest?

Slide 3: Big Question

3 min

Say: *"17% gone. Tipping point at 20-25%. That's a margin of 3-8%. At current rates, we have 15-25 years. Can we fix this?"*

Watch For

Students calculate: At 10,000 sq km per year of deforestation, how many years until 20%? Until 25%? Discuss the urgency.

Slide 4: LEVER Framework

2 min

Say: *"We're racing the clock. Build the model that tells us whether restoration can outpace destruction."*

Watch For

Preview each LEVER step. Emphasize that this is about timescales — destruction is fast, recovery is slow.

Slide 5: Activity 1: Components

8 min

Say: *"Two things we control directly: how fast we cut and how fast we plant. Everything else the ecosystem has to do on its own."*

Watch For

Guide sorting. Discuss why Deforestation Rate and Reforestation Effort are external (policy-driven) while ecological recovery processes are internal.

Slide 6: Activity 2: Connections

8 min

Say: *"You can plant a million trees tomorrow. But can those trees survive in depleted soil? Can they restart the rain cycle? How long does each step take?"*

Watch For

Help students trace the recovery cascade. Identify the bottleneck: Soil Nutrient Recovery is the rate-limiting step for everything downstream.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's test three futures for the Amazon. Which one saves it? Which one loses it forever?"*

Watch For

Students predict outcomes first, then run all three scenarios. Focus on the tipping point threshold in each. Discuss the 'partial intervention' scenario — is halfway enough?

Slide 8: Discoveries

5 min

Say: *"The model reveals an uncomfortable truth: even aggressive restoration takes decades to centuries. But the alternative is losing the Amazon entirely."*

Watch For

Lead discussion on the timescale mismatch. Compare destruction speed (years) to recovery speed (decades-centuries). Discuss what 'success' looks like at different timescales.

Slide 9: STEM Challenge

12 min

Say: *"Brazil just gave your team 10,000 square kilometers and 50 years. Design the plan that brings the Amazon back to life."*

Watch For

Groups design comprehensive restoration plans including species selection, phasing, monitoring milestones, and community engagement. Present as professional proposals.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Deforestation Rate, Reforestation Effort
- **INTERNAL:** Soil Nutrient Recovery, Water Cycle Restoration, Biodiversity Return, Carbon Sequestration, Tipping Point Threshold

Relationships

- **Deforestation Rate to Tipping Point Threshold:** POSITIVE (+) — Higher Deforestation Rate directly pushes the system closer to the critical threshold by removing the trees that generate rainfall and maintain the self-sustaining water cycle. Each hectare cleared reduces the forest's ability to sustain itself.
- **Reforestation Effort to Soil Nutrient Recovery:** POSITIVE (+) — Greater Reforestation Effort — planting native species in degraded areas — initiates the slow process of rebuilding soil organic matter, mycorrhizal networks, and nutrient cycling capacity that were destroyed by deforestation.
- **Water Cycle Restoration to Biodiversity Return:** POSITIVE (+) — As the water cycle recovers through increased evapotranspiration from maturing reforested areas, consistent rainfall supports the return of moisture-dependent species and enables ecological succession toward complex forest community structure.

Key Discoveries

- The Amazon's tipping point is dangerously close — approximately 17% of the original forest has already been cleared, with the threshold estimated at 20-25%
- Reforestation is not simply planting trees — Soil Nutrient Recovery takes 15-50 years and determines whether planted trees can actually survive and establish a self-sustaining ecosystem
- The water cycle is the critical feedback: the Amazon generates 50% of its own rainfall through evapotranspiration, so losing too much forest starves the remaining forest of water
- Full ecosystem recovery including Biodiversity Return of specialist species requires 100-200+ years — there is no quick fix for complex tropical forest ecosystems

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Planting any trees counts as restoration"	Ecologically meaningful restoration requires planting native species in appropriate combinations that can support the full web of ecological interactions. Monoculture plantations of eucalyptus or pine do not restore rainforest function — they lack biodiversity, do not support native wildlife, deplete soil differently, and do not generate the same evapotranspiration patterns. True restoration mimics natural forest composition.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Forests regrow quickly if you leave them alone"	While natural regeneration does occur in small clearings near intact forest, large-scale deforested areas face multiple barriers: depleted soils, invasive grasses that prevent tree seedling establishment, absence of seed dispersers, and altered microclimate (too hot and dry for forest species). In heavily degraded areas, active restoration is essential — natural regeneration alone could take 500+ years or may never occur.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"The Amazon is too big to lose"	The Amazon's size creates the illusion of invulnerability, but the tipping point model shows that the forest's survival depends on a self-sustaining feedback loop (evapotranspiration-rainfall) that has a critical threshold. Once 20-25% is cleared, the remaining 75-80% may not receive enough self-generated rainfall to survive. Size provides no protection against a system-level collapse.